Thoughts And Rambles On Probability And Statistics

Problem Set 5: Integral Transforms and Convolutions

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These are the end of chapter problems for Chapter 5 of the notes. Equation numbers refer to the full notes.

1. Let $X_1, X_2, \ldots, X_n$ be independent random variables with expectation 0 and finite third moments. Show, with the aid of characteristic functions, that

$$\mathbb{E}\left(X_1 + X_2 + \cdots + X_n\right)^3 = \mathbb{E}X_1^3 + \mathbb{E}X_2^3 + \cdots + \mathbb{E}X_n^3$$

2. Which of these cannot be characteristic functions? Explain.

1). $\cos(t)$

2). e^{t^2}

3). 1

3. Let $X \sim N(0,1)$ and let $Y = e^X$. Find $\mathbb{E}(Y)$ and $\mathrm{Var}(Y)$ (A repeated problem, but now with the characteristic function)

4. Prove that

$$\int_{-\infty}^{\infty} \frac{1}{\left(1+y^2\right)^2}\,\mathrm{d}y = \frac{\pi}{2},$$
$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{1-\cos y}{y^2}\mathrm{e}^{-|y|}\mathrm{d}y = \frac{1}{2} - \frac{\log 2}{\pi}.$$

5. Let X be a random variable, such that

$$EX^n = \frac{2n!}{3\lambda^n}, \quad n = 1, 2, \ldots$$

Find the distribution of X. Is it unique?

6. (Computational) Let $X_1, X_2, ..., X_n$ be i.i.d. geometric random variables. Using the probability generating function, plot the distribution of $\sum_{i=1}^n X_1 + X_2 + ... + X_n$ by calculating the probabilities.

7. Let X be an exponential random variable with $\lambda = 1$ and Y be a Poisson random variable

with $\lambda = 1$. Using the convolution, find the distribution of $X + Y$

8. Find the characteristic function of the normal distribution with mean μ and variance σ^2. Once you have the characteristic function, take the log to find the cumulant generating function. Find the first 4-5 cumulants. What do you notice? Generalize your results.

9. Given the function

$$e^{-c|t|^\alpha}$$

Where $c > 0$ is some constant and α takes on positive integer values. For what values of α can the above be a characteristic function?